TRENDY TECH

UPLIFT YOUR CAREER!

Azure Data Engine

Masters Prog

Azure Data Engineerir

Azure Data Engineering Fundamentals

Cloud and On-Premise

Characteristics of Cloud

IAAS, PAAS, SAAS

Cloud Deployment Models- Public, Private, Hybrid

Azure Microsoft Services

Resource, Resource Group, Subscription

Data Center

Azure Regions

Azure Availability Zones

Zonal Services & Zone-Redundant Services

Handling Datacenter Failures

Create and Manage Multiple VMs
Load Balancer
Availability Set Vs Availability Zone
Fault Domain & Update Domain

Azure Data Storage

Storage Account - Blob, Table, File, Queue services
Access Tiers -Data Accessibility
Locally Redundant Storage(LRS)
Zone Redundant Storage(ZRS)
Geo Redundant Storage(GRS)
Read Access Geo Redundant Storage
Geo-Zone Redundant Storage
Read Access Geo-Zone Redundant Storage
Introduction to DataLake

Azure Data Factory

Overview of Different Azure Services

Complete End to End Datapipeline - Usecase

Data Ingestion

Azure Data Factory Introduction

Data Transfer (Source to Sink)

Data Transformation - Data Flow

Workflow Orchestration

Data Transfer from RDBMS to ADLS Gen2

Azure SQL Databses

Data Transfer from Azure SQL to ADLS Gen2

Author, Monitor & Manage

Data Integration Service (ADF)

Usecases where ADF can be used

Data Ingestion

Linked Services for Source and Sink
Select Transformation
Practice Assignments
ADF Primary Usage
Tansfer data from Blob to Datalake - Usecase
Blob Connector
Http Connector
Datalake Instance
Data Factory Instance
Linked Service Creation - Blob & Datalake
Dataset for Blob and Datalake
Complete Pipeline setup
Pipeline Parameterization
Key Vault
Scheduled Triggers
Tumbling Window Triggers

Azure Synapse Analytics

Need for Datawarehouse

Best tool for Data Analysis

What is Azure Synapse Analytics

Synapse Usecase

Ingestion - Synapse Pipeline & Mapping Dataflow

Computation - SQL pools & Mapping Dataflow

Dedicated SQL Pool

Serverless SQL Pool

Apache Spark Pool

External Table & Normal Table

How to Create External Tables

Usecase for Serverless SQL Pool

OPENROWSET & External Table

2 ways to query data in ADLS Gen2

When to use Dedicated SQL Pool
Datawarehouse Architecture
Datawarehousing Units (DWU)
Fact and Dimension Tables
Table Distributions - Round Robin, Hash, Replicate
2 Options to Load Data from Datalake to Dedicated SQL
Polybase
Copy Command
Distribution Types in detail
Processing the data in dedicated SQL Pool with Spark
Processing the spark table through serverless SQL Pool
Azure Synapse Summary
Serverless SQL Pool - Openrowset, external tables
Dedicated SQL Pool - Control, Compute, Distribution, Po
Spark SQL Pool - Spark Tables, Dedicated SQL

Azure Databricks

What is Databricks

Why Databricks

Databricks Pricing - Infrastructure and Software Charge

Different Cloud Providers offering Databricks

Databricks Features

3 ways to create cluster

All Purpose Cluster

Job Cluster

Cluster Pool

Cluster Modes - Single Node, Standard, High Concurren

When to use the Different Cluster Modes

Databricks Benefits

Different optimized Cluster types

Memory Optimized, Storage Optimized, Compute Optim

Object Store - Blob, Datalake Gen2
Filesystem utility- dbutils
Data Utility
Notebook Utility
Widgets Utility
Parameter passing from one Notebook to another
Mount Point
How to create Mount Point
Databricks Workspace
Databricks CLI
Ways to access Storage Account
Access Key/ Account Key
SAS Key
Service Principal
Secret Scope
Azure Key vault Backed Secret Scope

Azure Delta Lake

What is Datalake

Advantages and Challenges of Datalake

Example Usecases

Why Delta Lake

Updates and Deletes in Delta Lake

Azure Portal Setup

Cluster creation in Databricks

Delta Table Creation

Inserting values into Delta Table - Insert, Append, Copy

Schema Evolution

Azure Lakehouse Architecture

Datalake Benefits and Challenges

2-tier modern Datawarehouse Architecture

Azure Delta Engine

Delta Cache

2 ways to enable Delta Cache

Optimizations

Data Skipping using Stats

Delta Cache

Small File Problem

Compaction/Bin-Packing

Optimize Command

Z-Ordering

Partitioning Bucketing

Vacuum Command

Auto Optimization and Auto Compaction

Optimized Writes

Photon Query Engine

1 End to End Real-Time Projects

(Sumit Sir Will be open to add more topics as per the Indus

Practice Environment

-Create your own Azure Account and you could use the Free C
you will be charged a nominal amount for the rest of your pract
-First few modules like Fundamentals, Data Storage and Data I
most part of Databricks can be practiced on community editio
-Synapse Service has to be used cautiously as it is a slightly e
as explained by Sumit Sir in the videos, you will not be charg
practice)

Important Note : Please note that the course validity is 2 years

Thank Yo

Contact : hello@tr